

# ROCm 7: AMD's big open-source bet on the future of AI

ROCm 7 brings AMD's open AI push to the forefront with faster LLMs, Windows support, and Radeon GPU compatibility.

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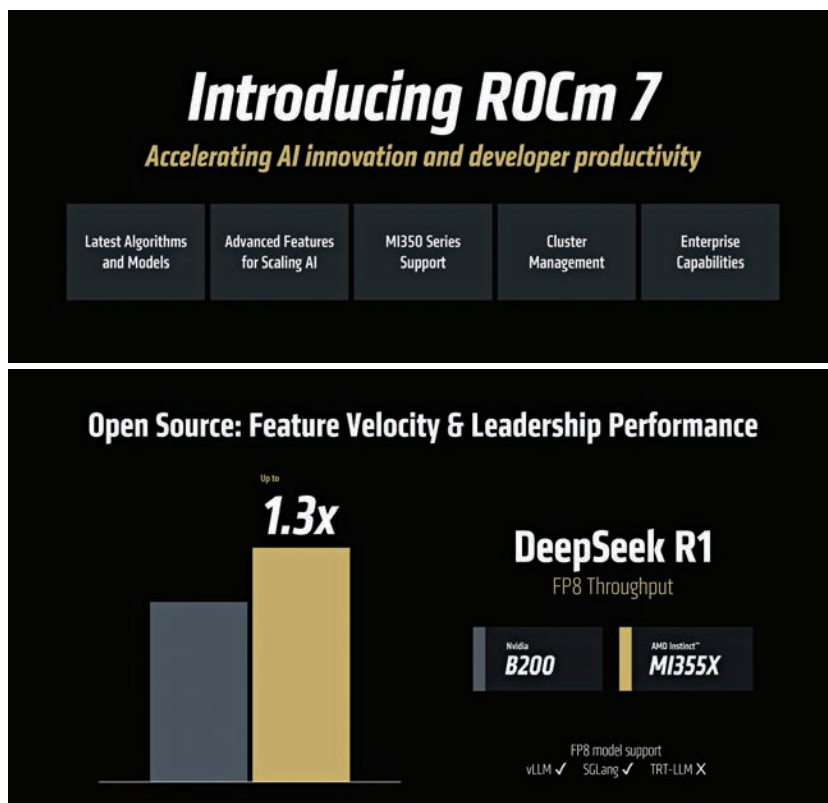
**I**f 2023 and 2024 were the years NVIDIA set the pace for AI acceleration, 2025 is shaping up to be the year AMD answers back with more than just hardware. At its flagship "Advancing AI" event, AMD not only launched the Instinct MI350 series accelerators but also introduced ROCm 7, its latest iteration of the open AI software platform that's quietly become a centrepiece of its AI ambitions.

While silicon often steals the lime-light, software is where most of the AI magic actually happens. And for AMD, ROCm 7 is a substantial evolution that aims to close the gap with NVIDIA's CUDA ecosystem, expand AI software support to a wider range of devices, and finally position AMD as a true end-to-end AI player, from cloud datacentres all the way to client PCs.

## What is ROCm 7?

ROCm, short for Radeon Open Compute, is AMD's open-source software platform designed to enable high-performance computing (HPC) and AI workloads on its GPUs. Launched in 2016, it was aimed at data scientists and developers working on scientific simulations and neural networks.

With ROCm 7, AMD is clearly pivoting towards a full-stack AI platform that supports both training and inference at scale. And this isn't just a tweak under the hood. ROCm 7 brings a generational leap, supporting newer low-precision formats, enabling better distributed computing, and rolling out



for the first time with Windows support. Yes, including WSL.

From improved developer tools to new model-serving frameworks, ROCm 7 represents a significant effort to open up AI development to more platforms and more people.

## What's new in ROCm 7?

Performance uplift: At the heart of ROCm 7's improvements is raw speed. AMD claims up to 3.5× improvement in inference workloads and 3× gains in training performance compared to

ROCm 6. That's not just theoretical marketing fluff, the company showcased several benchmarked models including Meta's Llama 3.1-70B, Alibaba's Qwen2-72B, and DeepSeek R1. For developers working with transformer-heavy LLMs, these aren't minor boosts. The optimisations in ROCm 7 come from both better memory utilisation and new support for low-precision data types, which can significantly reduce compute overhead.

Support for FP4 and FP6 Low-Precision Formats: One of the marquee